Summary in Graph

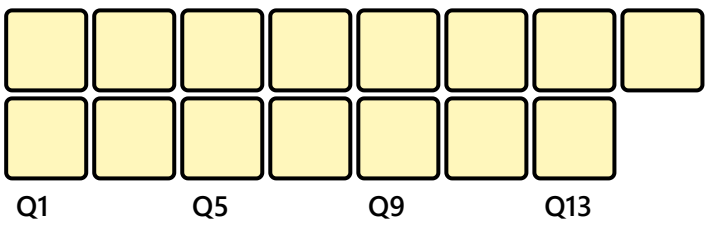
Exam Summary (GO Classes GATE CS | Operating System | File Systems | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (15 + 0)
Total Marks:	15 (15 + 0)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (14) out of 135 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Operating System

Consider a file system that uses **contiguous allocation**.

A file of size **100 blocks** is already stored contiguously on disk.

A new block must be inserted at the **beginning** of the file, but the disk block at the start of the file is **not**

free, so the file cannot remain at its current location.

Assume:

- The data of the new block is already in memory.
- The file control block (FCB) is already in memory.

Only the reads and writes of **actual file data blocks** are counted as disk accesses.
How many disk accesses (read + write) are required to perform this insertion?

A. 100
B. 200
C. 201
D. 101

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file system that uses **linked allocation**.
A file containing **100 blocks** is stored on disk.
A new block must be inserted at the **beginning** of the file.
Only actual block reads/writes count as disk accesses.

Assume:

- The file control block (FCB) is already in memory.
- New block's content is already in memory.

How many disk accesses (read+write) are required?

A. 1
B. 2
C. 100
D. 101

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file system using **single-level indexed allocation**.
A file consists of **100** blocks, and its index block is already in memory.
To insert a new block at the beginning, the index entries must shift in memory, and the new block must be written to disk.

How many disk accesses are required?

A. 1
B. 2
C. 100

D. 101

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file system that uses **contiguous allocation**.
A file of size **100 blocks** is already stored on disk.
A new block must be inserted between the $\mathbf{50}^{\text{th}}$ and $\mathbf{51}^{\text{st}}$ blocks.
Assume the file control block (FCB) is already in memory.

How many disk accesses are required?

A. 50

B. 100

C. 101

D. 150

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored using **linked allocation**.
The file contains **100 blocks**.
A new block is to be inserted after the $\mathbf{50}^{\text{th}}$ block.
Assume the file control block (FCB) is already in memory.
Let the number of disk accesses required be $\mathbf{n}$.

Which of the following is the correct value of $\mathbf{n}$?

A. 50

B. 51

C. 52

D. 100

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file system using **single-level indexed allocation**.
A file has **100 blocks**, and its index block is in memory.
To insert a block at position 50, pointer shifting happens entirely in memory.

How many disk accesses are required?

A. 1

B. 2

C. 100

D. 101

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored using **contiguous allocation**.
The file occupies **100 blocks**, and free space exists immediately after it.
A new block must be appended at the end.
The new block's data is already in memory.

How many disk accesses are required?

A. 1

B. 2

C. 100

D. 101

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored using **linked allocation**.
The file currently consists of **100 blocks**.

If you want to insert a new block at the **end** of the file, how many **disk accesses** will be required?

A. 100
B. 101
C. 102
D. 103

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored contiguously with **100 blocks**.
Removing the first block requires shifting the remaining **99 blocks** one position left.

How many disk accesses are required?

A. 198

B. 199

C. 200

D. 201

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A file stored using **linked allocation** has **100 blocks**.
We want to remove the $\mathbf{50}^{\text{th}}$ block.
Only actual data block reads/writes count as disk accesses.

How many disk accesses are required?

A. 50

B. 51

C. 52

D. 100

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored using **linked allocation**.
The file consists of **100 blocks**.
We want to **remove the first block** of the file.
Assume the directory entry points to the first block, and only actual block reads/writes count as disk accesses.

How many disk accesses are required?

A. 1

B. 2

C. 100

D. 101

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider a file stored using **single-level indexed allocation**.
The file contains **100 blocks**, and the **index block is already in memory**.
We want to **remove the first block** from the file.

How many disk accesses are required?

A. 0

B. 1

C. 50

D. 100

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A file is stored using **contiguous allocation** and occupies **100 blocks**.
Block positions are numbered from **0 to 99** (as usual).

We want to **remove the block at position 50**. To maintain contiguity, all blocks after it will be shifted one position to the left.
Only the **actual reads and writes of file data blocks** are counted as disk accesses.

How many disk accesses are required?

A. 49
B. 50
C. 98
D. 100

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A file stored using **linked allocation** contains **100 blocks**.
Block positions are numbered from **0 to 99**, as is standard.

We want to **remove the block at position 50**.
Only the **actual reads and writes of data blocks** are counted as disk accesses.

How many disk accesses are required?

A. 50
B. 51
C. 52
D. 100

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

A file stored using **linked allocation** contains **100 blocks**.
We want to **remove the last block** of the file.
Only disk block reads/writes count as disk accesses.

How many disk accesses are required?

A. 1

B. 99

C. 100

D. 101

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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